Roberto Nasrt Shawky

Junior Security Analyst

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SUMMARY

Mechatronics Engineering graduate with hands-on experience in control systems, industrial automation, and embedded systems. Completed intensive training in penetration testing and DevOps through the Digital Egypt Pioneers Initiative (DEPI). Ranked Top 6% on TryHackMe (100+ labs completed). Skilled in web and network security testing, vulnerability assessment, and exploitation, with a strong foundation in real-time systems and industrial control environments. Seeking entry-level roles in OT/ICS security, penetration testing, or security operations.

EDUCATION

B.Sc. in Mechatronics Engineering **Graduated July 2025**

Ain Shams University | Cairo, Egypt

Relevant coursework: Automatic Control Systems, Industrial Robotics, Industrial Automation, Hydraulic and Pneumatic control, Modeling and Simulation of Mechatronics, Mechatronics systems design, Computer programming, Embedded Software, Industrial Networks

Student Activities:

AERO ASU – Flight Control Engineer, Aviation Club – Robolympics Electrical Technical Support

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EXPERIENCE

Vulnerability Analyst and Penetration Tester (Training) **December 2025 – Present**

Digital Egypt Pioneers Initiative (DEPI)

- Analyzed modern web application architectures (Microservices, SPA, APIs, databases) to identify attack surfaces and perform white and black box testing
- Conducted network reconnaissance and exploitation using Nmap and Metasploit against home lab environments such as Metasploitable and Mr Robot (VulnHub VMs)
- Performed web application penetration tests using OffSec (OSWA) and INE (eWPTX) methodologies against a home lab (DVWA) and Online web CTFs
- Discovered, exploited and documented (notes and PoC) vulnerabilities including XSS, IDOR, SQLi, and authentication flaws in lab environments by manual testing

DevOps Engineer (Training) **April – October 2024**

Digital Egypt Pioneers Initiative (DEPI)

- Developed Infrastructure as Code (IaC) with Terraform for AWS resource provisioning
- Managed containerized applications with Docker and Kubernetes
- Built CI/CD pipelines using GitHub Actions, ensuring automated integration and delivery
- Automated configuration management using Ansible to maintain system reliability

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PROJECTS

CTF Challenges (TryHackMe / Hackviser) **October 2025 – Present**

- Solved real-world inspired web and network security challenges involving reconnaissance, enumeration, exploitation, and privilege escalation
- Documented attack paths and solutions for multiple labs, improving exploitation methodology and reporting skills
- Top 6% on TryHackMe, completed 100+ rooms in offensive security paths including Junior Pentester, Web Pentesting and Red teaming
- Completed all CTF challenges for Hackviser CAPT certification track

Active Suspension System – Graduation Project **September 2024 – June 2025**

- Led architecture design and overall tasks to reduce the suspension travel and acceleration
- Modeled and simulated a quarter-car suspension system using state-space and block diagrams
- Designed and implemented advanced controllers (LQR, Sliding Mode, Reinforcement Learning)
- Conducted Hardware-in-the-Loop (HIL) testing in MATLAB/Simulink to validate embedded control algorithms

Rotary Inverted Pendulum (LQR Control System) **September 2024 – June 2025**

- Designed and implemented a rotary inverted pendulum using ESP32 and MATLAB/Simulink
- Developed a state-space model and applied an LQR controller for system stabilization
- Integrated DC motor with encoder and pendulum angle encoder for real-time feedback
- Tuned Q and R matrices to achieve optimal balance and system response
- Implemented real-time control loop for stabilizing the pendulum in the upright position
- Validated system performance through simulation and hardware testing

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TECHNICAL SKILLS

Control Systems & Engineering: MATLAB/Simulink, State-Space Modeling, LQR Controllers, Sliding Mode Control, Hardware-in-the-Loop (HIL) testing

Industrial automation: Sensors, actuators, motor control, PLC, DCS, SCADA, HMI

Penetration Testing (Web): XSS, IDOR, CSRF, SSRF, SQL Injection, Command Injection

Methodology: Reconnaissance, enumeration, exploitation, reporting

Tools: Burp Suite, Nmap, ffuf, Gobuster & Metasploit

Operating Systems: Linux (Void, Mint, Ubuntu, Kali, Parrot) and Windows

DevOps Tools: Docker, Ansible, Terraform, Kubernetes, Git, GitHub

Practices: Agile (Scrum), planning and leading

CERTIFICATIONS & COURSES

Security and Penetration Testing:

TryHackMe – Web Application Pentesting
Hackviser - Certified Associate Penetration Tester
Sprints x Microsoft Summer Camp – Cybersecurity
Red Team Leaders - OpSec & Anonymity for Red Teams

Mechatronics:

Coursera - Machine Design Part I
Coursera - Sensor and Sensor Circuit Design
Coursera - Motors and Motor Control Circuits
Coursera - Introduction to Programming with MATLAB

Robotics:

Coursera - Building Arduino robots and devices
Coursera - Modern Robotics, Course 1: Foundations of Robot Motion

Embedded software:

Coursera - C Programming with Linux Specialization (7 courses)
Coursera - Embedded Software and Hardware Architecture
Learn in Depth - Master C/Embedded C for embedded system